

Airway and circulating levels of carotenoids in asthma and healthy controls.

BACKGROUND: Elevated oxidative stress and impaired antioxidant defences are increasingly recognised features of asthma. Carotenoids are potent dietary antioxidants that may protect against asthma by reducing oxidative damage. **OBJECTIVES:** This study aimed firstly, to characterise circulating and airway levels of carotenoids in asthma compared to healthy controls, in relation to dietary intake. Secondly, the study aimed to test whether airway lycopene defences can be improved using oral supplements. **METHODS:** Induced sputum and peripheral blood samples were collected from subjects with asthma ($n = 15$) and healthy controls ($n = 16$). Dietary carotenoid intakes were estimated using the 24-hour recall method and analysed using a modified version of the Foodworks 210 Nutrient Calculation Software. Another group of healthy controls ($n = 9$) were supplemented with 20 mg/day lycopene for 4 weeks. Carotenoids (beta-carotene, lycopene, alpha-carotene, beta-cryptoxanthin, lutein/zeaxanthin) were measured by HPLC. **RESULTS:** Despite similar dietary intake, whole blood levels of total carotenoids, lycopene, lutein, beta-cryptoxanthin, alpha-carotene and beta-carotene were significantly lower in asthma than controls. However, there were no differences in plasma or sputum carotenoid levels. Induced sputum carotenoid levels were significantly lower than plasma and whole blood levels, but correlated strongly with plasma levels ($r = 0.798$, $p < 0.001$). Although there were no overall increases in either plasma or sputum lycopene levels following supplementation, changes in airway lycopene levels correlated with changes in plasma levels ($r = 0.908$, $p < 0.002$). **CONCLUSIONS:** Whole blood, but not plasma or sputum, carotenoid levels are deficient in asthma. Plasma carotenoid levels reflect airway carotenoid levels and when plasma levels are improved using oral supplements this is reflected in the airways.